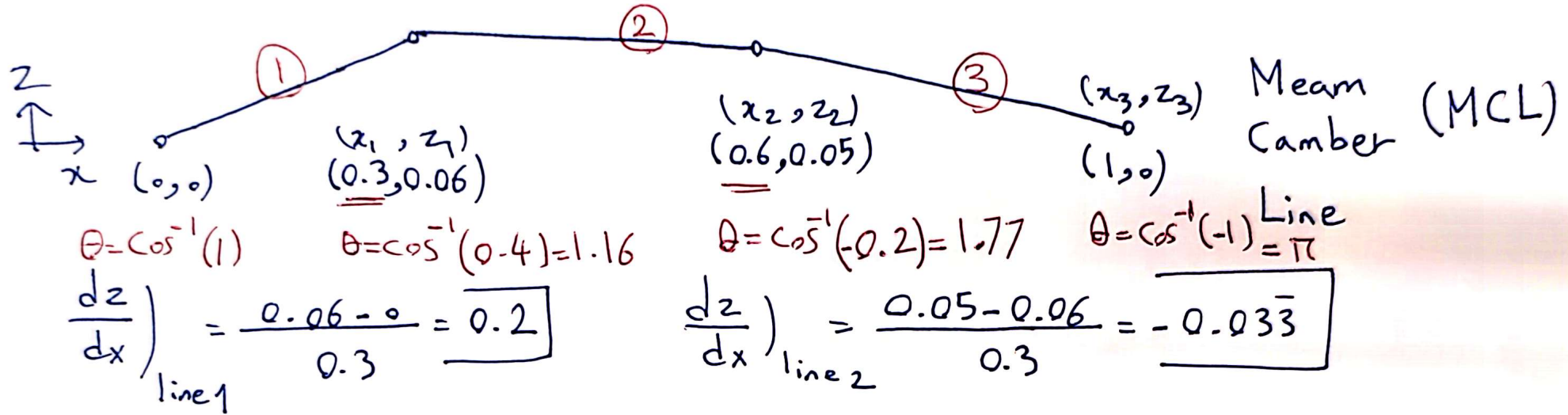


is this maximum camber y , and where is it location x ?

2. For a NACA 6310 airfoil, visit the Airfoil Tools website (<http://airfoiltools.com>) and click on "NACA 4 digit airfoil generator" to visualize this shape. Print this out so as to draw in a simplified MCL using three (3) straight line segments: (0, 0) to (0.30, 0.06), (0.30, 0.06) to (0.60, 0.05) and (0.60, 0.05) to (1, 0). Then, using thin-airfoil theory:
- Determine the Fourier Coefficients A_0 , A_1 and A_2
 - For $\alpha = 0$, 5 and 10 degrees, determine C_l , $\alpha_{L=0}$ and $C_{m_{c/4}}$ at each AoA. Plot this to a curve.



$$\left. \frac{dz}{dx} \right|_{\text{line 3}} = \frac{0 - 0.05}{1 - 0.6} = -0.125$$

Transformation from $x(\xi)$ to

$$\theta \rightarrow \xi = \frac{1 - \cos \theta}{2}$$

$$\text{so } \cos \theta = 1 - 2\xi \rightarrow \theta = \cos^{-1}(1 - 2\xi)$$

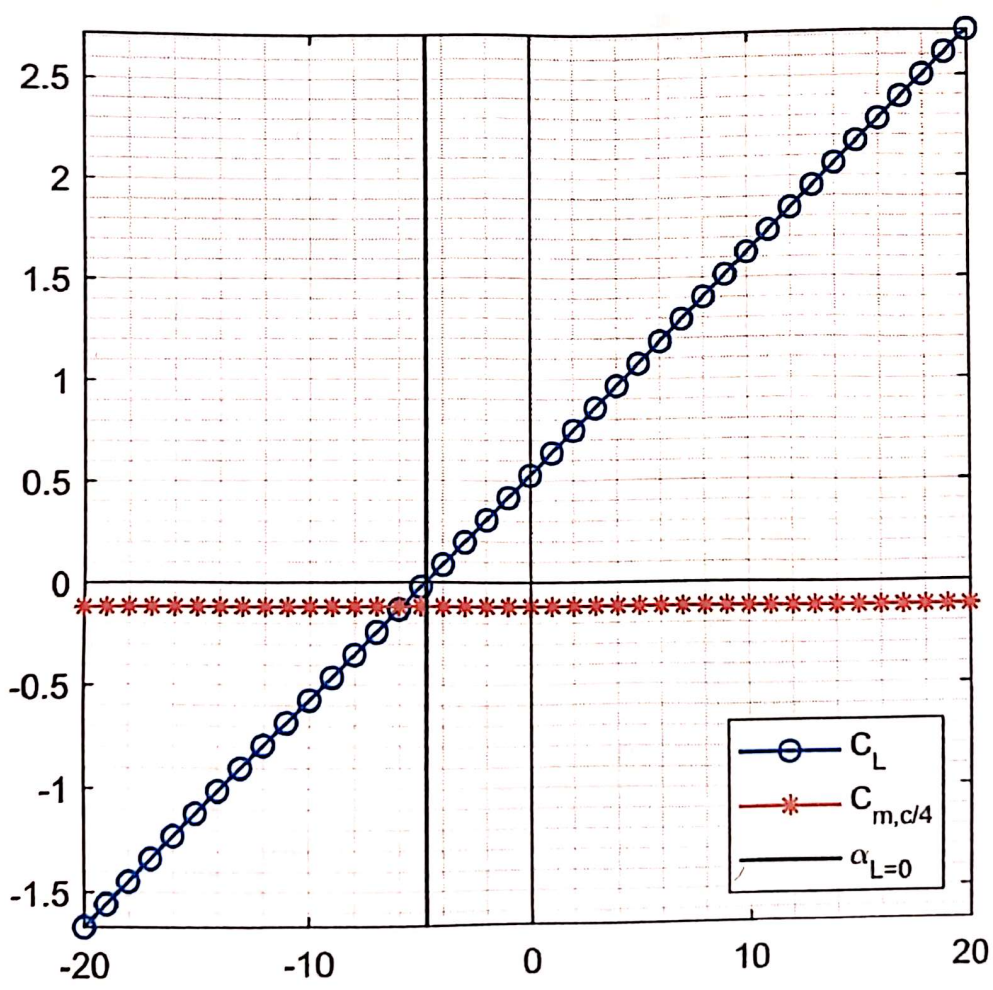
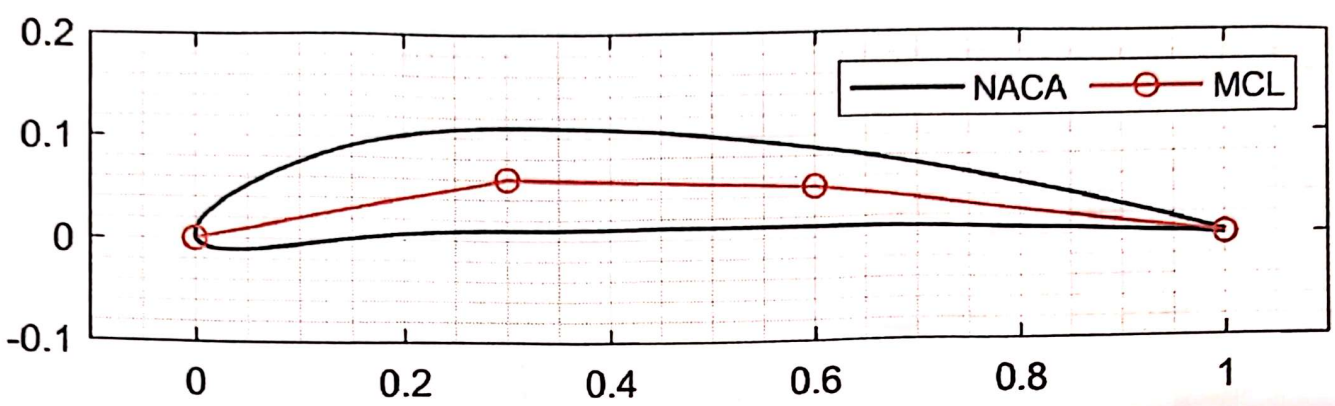
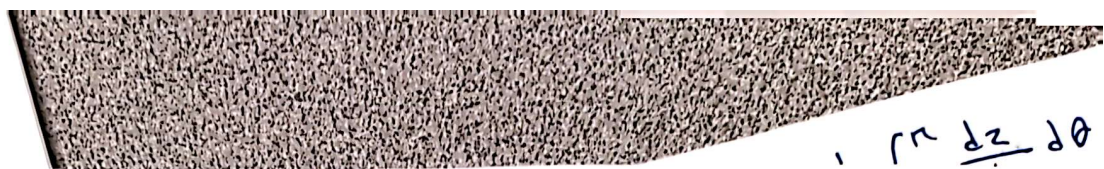
Everything should be in radians

$$A_o = \alpha - \frac{1}{\pi} \int_0^{\pi} \frac{dz}{dx} d\theta = \alpha - \frac{1}{\pi} \left[\int_0^{1.16} 0.2 d\theta + \int_{1.16}^{1.77} -0.033 d\theta + \int_{1.77}^{\pi} -0.125 d\theta \right]$$

$$A_o = \alpha - 0.012$$

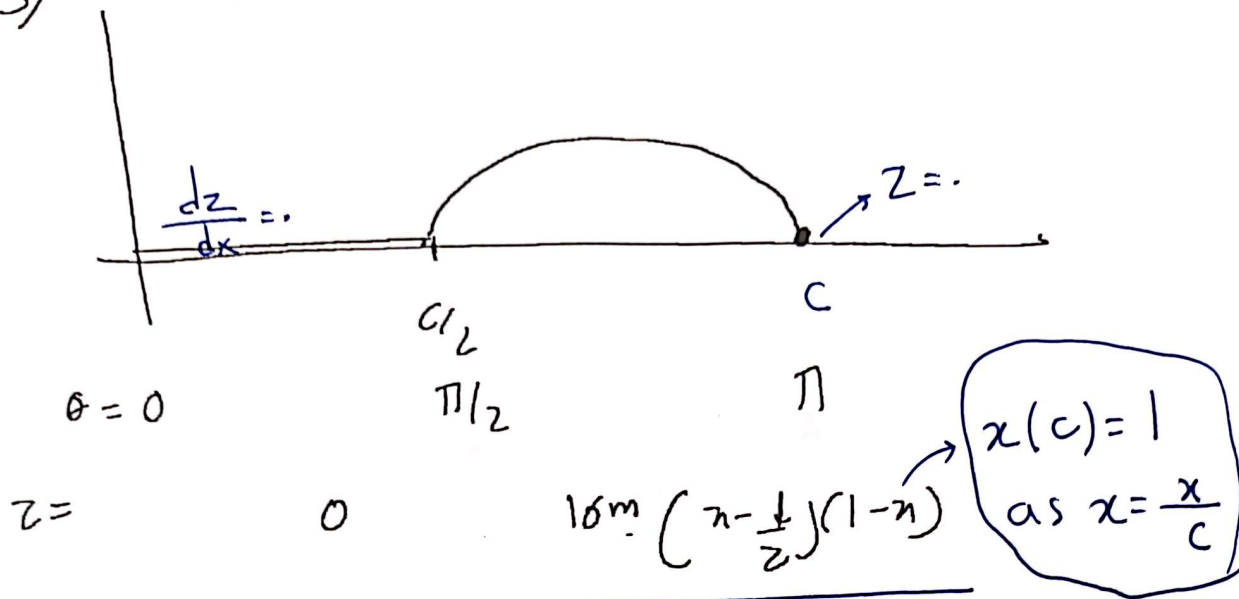
0838 496

$\frac{1}{r} \frac{dz}{d\theta}$



Q3)

Substrate: $C=1$



$$C_L = 2\pi \left(\alpha + \frac{1}{\pi} \int_0^\eta \frac{dz}{dx} (\cos \theta_0 - 1) d\theta_0 \right)$$

ve $\eta = \frac{1}{2} (1 - \cos \theta)$

$$\frac{dz}{dx} = z' = 0.96 - 1.28 \times \frac{1}{2} (1 - \cos \theta)$$

$$C_1 = 2\pi \left(a + \frac{1}{n} \int_0^{n/2} z' (\cos \theta - 1) d\theta \right)$$

$$= 211 (\alpha + 0.101859)$$

$$\therefore \alpha_{V=0} = -0.16859 \text{ rad}$$

$$C_{b, \alpha=30} = 0.968987$$

$$z = 16m(x - \frac{1}{2})(1-x), \quad x = \frac{x}{c}$$

$$\begin{aligned} \frac{dz}{dx} &= 16m(1-x) - 16m(x - \frac{1}{2}) \\ &= \cancel{16m} - \cancel{16mx} - \cancel{16mx} + \cancel{8m} \\ 24\cancel{m} - 32mx &= \underline{8m(3-4x)} \end{aligned}$$

$$C_l = 2\pi \left(\alpha + \frac{1}{\pi} \left[\int_{\frac{\pi}{2}}^{\pi} 0 + \int_{\frac{\pi}{2}}^{\pi} \underline{8m(3-4x)(\cos\theta-1)} d\theta \right] \right)$$

↓
Should be
converted to θ

$$3-4x = 3-4 \frac{1-\cos\theta}{2} = 3-2(1-\cos\theta) = \underline{1+2\cos\theta}$$

$$\rightarrow C_l = 2\pi \left(\alpha + \frac{1}{\pi} \int_{\frac{\pi}{2}}^{\pi} \underline{-8m(1+2\cos\theta)(1-\cos\theta)} d\theta \right)$$

$$C_l (\alpha=3^\circ) = 0.96987$$